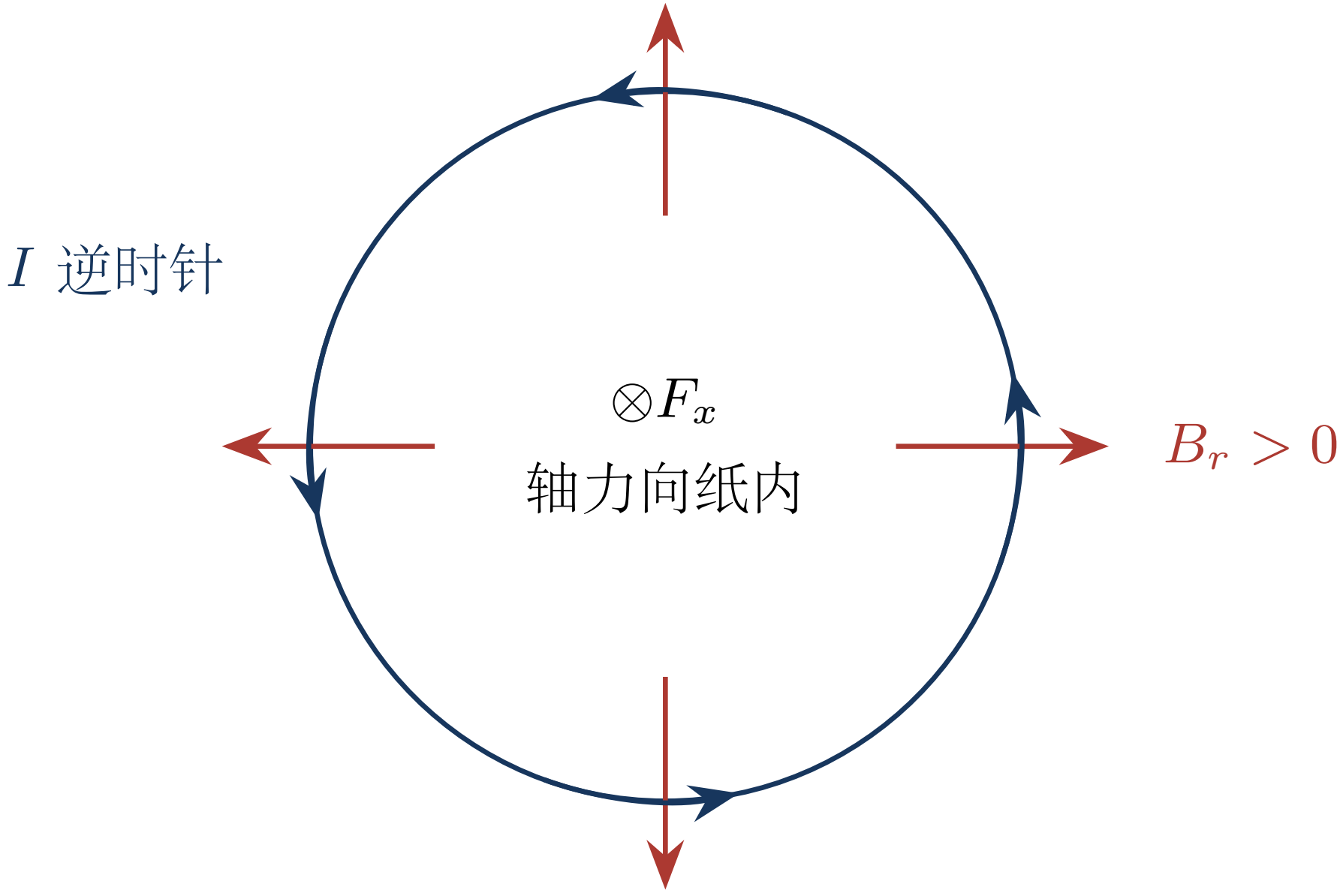

以 $+x$ 朝向观察者，$\mathbf{e}_\varphi \times \mathbf{e}_r = -\mathbf{e}_x$